microsoft/**Fara1.5-4B**

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Model card | Files xet | Community 2

Fara1.5-4B

Microsoft Project | Github | Paper 2606.20785

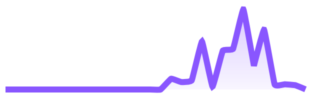
Azure Foundry

Fara1.5-4B is a multimodal **computer use agent (CUA)** for web browsers, from **Microsoft Research AI Frontiers**. It observes the browser through screenshots and acts on the user's behalf by emitting structured tool calls — click, type, scroll, visit URL, web search, and so on — to complete tasks end-to-end.

The model is vision-only at perception time: it sees the browser through screenshots, not the DOM or accessibility tree. Internal reasoning and trajectory history are tracked as text. Given the latest screenshot and prior actions, it predicts the next action with grounded arguments (e.g., pixel coordinates for a click).

Fara1.5-4B is supervised fine-tuned from Qwen3.5-4B on data generated by **FaraGen1.5**, our multi-

Downloads last month
4,717



Safetensors

Model size | 5B params

Tensor type | BF16

Chat template

Files info

Inference Providers NEW

Image-Text-to-Text

This model isn't deployed by any Inference Provider.

Ask for provider support

Model tree for microsoft/Fara1.5...

Base model Qwen/Qwen3.5-4B-Base

Finetuned Qwen/Qwen3.5-4B

Finetuned (435) this model

Finetunes 1 model

Quantizations 9 models

agent pipeline that synthesizes web tasks, executes trajectories to solve them, and verifies the results before training.

Please use only with the harness developped in Github/fara CLI or Magentic-Lite.

It's co-designed with **MagenticLite**, and that's the recommended deployment for both research and production.

Highlights

- **End-to-end web task completion.** Fills forms, books reservations, applies for jobs, plans trips, runs shopping carts. Not just clicking around — sequencing actions toward a goal.
- **Vision-only perception.** Operates on screenshots alone, no DOM access required. Matches the input modality available to a human user.
- **Coordinate-grounded actions.** Predicts pixel-level click and drag targets directly. No separate grounding model needed.
- **Critical-points safety design.** Trained to stop and ask before personal info entry, payments, submissions, sign-ins, sending messages, or other irreversible actions — even if it could technically continue.
- **262K context.** Long enough for multi-screenshot trajectories with full action history.
- **On-device viable.** 4B parameters means it runs on a single A100/H100/B200 with room for the screenshot history.

Model Details

Developer	Microsoft Research AI Frontiers
Architecture	Multimodal decoder-only LM (image + text → text)
Parameters	4B
Context length	262,144 tokens
Inputs	User goal (text), current screenshot(s), prior agent thoughts and actions
Outputs	Chain-of-thought block followed by a tool-call block (XML-tagged)
Training period	January 2026 – April 2026
Training compute	32 × NVIDIA B200, 2 days
Release date	21 May 2026
License	MIT
Base model	Qwen3.5-4B

Recommended Deployment: MagenticLite

The safest way to run Fara1.5-4B is inside **MagenticLite**, which provides:

- **Sandboxing** — the browser runs in a Docker container with no access to host files or environment variables

- **Allow-lists** — restrict navigation to a user-specified set of domains
- **Watch-mode** — real-time monitoring of every action with full trace logs
- **Pause** — immediate halt of agent activity at any point

If you integrate Fara1.5-4B directly, you're responsible for these controls. Don't run this model with unrestricted browser access on a machine that has anything sensitive on it.

Quickstart

Requirements

- `torch >= 2.11.0`
- `transformers >= 5.2.0`
- `vllm >= 0.19.1`
- A GPU with enough memory for a 4B model in bf16 (A6000, A100, H100, and B200 have been tested)

Serve with vLLM

```
vllm serve microsoft/Fara1.5-4B \
  --dtype bfloat16 \
  --max-model-len 262144 \
  --limit-mm-per-prompt image=10
```

System prompt

Fara1.5-4B is trained against a specific system prompt. Use it verbatim for best results:

```
You are Fara, a computer use agent (CUA) spe

The model was trained in the timeframe of Ja

This edition of the model was trained using 

A critical point is a situation where we mus

Case 1: Missing User Information — The task 

Case 2: Underspecified Task — The task descr

Case 3: Irreversible Action — We are about t

Only stop at a critical point if (1) require
```

The full system prompt, including the complete `computer_use` tool schema, ships with the model in MagenticLite.

Tool schema

Fara1.5-4B emits actions as `<tool_call>...</tool_call>` XML blocks containing a JSON object that calls the `computer_use` function. Supported actions:

Action	Purpose
`left_click`, `right_click`, `double_click`, `triple_click`	Mouse clicks at `(x, y)`
`mouse_move`, `left_click_drag`	Cursor positioning and drag
`type`, `key`	Keyboard input
`scroll`, `hscroll`	Page scrolling

Action	Purpose
visit_url, history_back	Browser navigation
web_search	Search query
pause_and_memorize_fact	Persist a fact across the trajectory
ask_user_question	Surface a clarifying question to the user
wait	Sleep for N seconds
terminate	End the task with a final answer

The screen resolution Fara is most commonly trained with is **1440×900**. Match this in your sandbox for the most reliable grounding.

Minimal agent loop

Note: We strongly recommend to use only with the harness developped in Github/fara or Magentic-Lite.

```
# Pseudocode for a single-step interaction.
# endpoint and pass the screenshot as an ima

screenshot_b64 = capture_browser_screenshot(

messages = [
    {"role": "system", "content": FARA_SYSTE
```

```
    {"role": "user", "content": [
        {"type": "text", "text": "Book a tab
        {"type": "image_url", "image_url": {
    ]},
]

response = client.chat.completions.create(
    model="microsoft/Fara1.5-4B",
    messages=messages,
    temperature=0.0,
    max_tokens=2048,
)

# Parse the assistant's reply: it will conta
# followed by <tool_call>{"name": "computer_
# Execute the action in your sandboxed brows
# append the assistant turn and a tool/obser
# the model emits action="terminate". We onl
# in the chat history.
```

A reference implementation of the full agent loop, including screenshot capture and Docker sandboxing, is in MagenticLite.

Critical Points: Safety by Design

Fara1.5-4B is trained to **pause and ask the user** at three types of critical points:

1. **Missing user information.** If the task needs personal data (name, email, phone, address, payment) the user hasn't provided, fill in what's available and stop to ask for the rest. Never fabricate.
2. **Underspecified tasks.** If the goal is ambiguous at the current decision point (e.g., "book me a flight" with no destination), stop and clarify.

3. **Irreversible actions.** Submitting forms, completing purchases, sending messages, deleting data — stop unless the user explicitly authorized the action upfront.

Concrete actions Fara is trained to stop on without explicit authorization:

- Entering personal information (name, email, phone)
- Entering credit card or shipping/billing details
- Completing purchases or bookings
- Making phone calls
- Sending emails
- Submitting job applications
- Signing into accounts

If the user grants authorization, the model proceeds. The principle is that costly, irreversible actions require a human in the loop.

Evaluation

We evaluate Fara1.5-4B on end-to-end web agent benchmarks. For WebTailBench, we report outcome success.

Model	WebVoyager	Online-Mind2Web	WebTailBench
Fara1.5-4B	80.8	57.3	27.4
Fara1.5-9B	86.6	63.4	32.3

Model	WebVoyager	Online-Mind2Web	WebTailBench
Fara1.5-27B	89.3	72.3	40.2

Training

Approach

Supervised Fine-Tuning on top of Qwen3.5-4B. Training mix is generated by FaraGen1.5, our multi-agent data pipeline, supplemented with curated public datasets for grounding and UI understanding.

Data sources

- **Synthetic trajectories** — Tasks seeded from URLs sampled from a large web index and from open-source seed datasets (e.g., Mind2Web). A multi-agent system attempts each task while recording screenshots, thoughts, and actions. A verifier agent then keeps only successful trajectories.
- **Grounding** — Curated datasets for predicting actions and pixel coordinates from screenshots, with images, text, and bounding boxes.
- **UI understanding** — Visual question answering, captioning, and OCR over web page screenshots collected by the pipeline.
- **Safety and instruction following** — Refusal data covering harmful or unsafe tasks the model should decline.

Scale

- Approximately 1 billion text tokens
- Less than 1 billion training images
- Latest data acquisition: March 20, 2026
- Training start: January 2026

The corpus is static. Future updates will ship as separately versioned models with their own model cards.

Intended Use and Limitations

Primary use cases

Automating repetitive web tasks: filling forms, shopping, booking travel, restaurant reservations, information seeking, account workflows. Fara1.5-4B can also serve as a grounding model for other agents that need pixel-accurate action prediction.

Out of scope

- Languages other than English (training data is English-only)
- High-stakes domains (legal, health, financial advice) where inaccurate actions could cause harm
- Allocation decisions affecting legal status, housing, employment, or credit
- Unsandboxed deployments with access to sensitive accounts or files
- Commercial or real-world production use without additional testing and safeguards

Known limitations

- **Vision-only perception** means the model can be misled by deceptive or low-quality page rendering, prompt injections embedded in page content, or visual ambiguity in UI elements
- **Multi-step trajectories accumulate error** — a misclick early in a sequence can compound
- **Run-to-run variance** on multi-turn tasks is non-trivial; benchmark numbers are averaged over multiple runs
- The model can hallucinate page state or misattribute information from earlier screenshots

Responsible AI Considerations

Computer use is a powerful capability. An agent that can click, type, and submit in a real browser can also do those things wrong. We strongly recommend:

- **Human-in-the-loop monitoring** of Fara's actions on the live web with a fast way to halt execution
- **Sandboxed execution** — run Fara in an isolated container with no access to host files, environment variables, or sensitive credentials
- **Allow-listed or block-listed browsing** — restrict the agent's reachable surface to limit exposure to malicious pages
- **No credential or PII sharing** with the agent unless absolutely required and the user has

authorized it

- **Output verification** — Fara can hallucinate, misattribute sources, or be misled by deceptive content; verify before acting on its outputs

Safety evaluation

Fara1.5-4B was evaluated via automated red-teaming on Azure. Coverage included groundedness, jailbreak resistance, harmful content (hate, violence, self-harm), and copyright violations. Safety post-training includes both refusal data for malicious tasks and the critical-points framework described above.

Known model risks

Like all language models, Fara1.5-4B can produce unfair, unreliable, or offensive outputs. Specific concerns:

- **Quality of service** varies across English dialects and is worse for non-English content
- **Representation harms** may persist despite safety post-training due to base-model biases
- **Information reliability** — generated content may be inaccurate or outdated
- **Prompt injection** — Fara reads page content, and adversarial content on a page may attempt to redirect its behavior

Developers deploying Fara1.5-4B should apply responsible AI best practices and ensure compliance with applicable laws and regulations.

Using safety services like Azure AI Content Safety is recommended.

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Authorized representative: Microsoft Ireland Operations Limited, 70 Sir John Rogerson's Quay, Dublin 2, D02 R296, Ireland.